

Welcome to Jupiter: The Giant That Could Have Been a Star!

From: NG Kids -- Space Mined: 2026-05-11 3 min read

Imagine a planet so huge it could hold 1,300 Earths inside it! That's Jupiter, the biggest planet in our solar system. Some scientists even call it a failed star. But don't worryyou can't land on it because Jupiter is a gas giant, made mostly of swirling **hydrogen** and **helium**. If you tried to skydive, you'd sink into thick clouds until the **pressure** turns the gas into a hot liquid. Yikes!

Jupiter has wild weather. Its clouds whip around at over 300 miles per hour, and the Great Red Spot is a giant hurricane that's been blowing for hundreds of years. It's so big that nearly three Earths could fit across it!

Jupiter also has almost 70 moons. One moon, Io, has volcanoes that shoot **sulfur** 300 miles high. Another, **Europa**, is covered in ice, but scientists think a huge ocean hides underneathand maybe even alien life!

But watch out: Jupiter's **radiation** is super dangerous, more than a thousand times what would be deadly to humans. So we'll just explore from our spaceship, sipping hot cocoa and watching the amazing light show.

Word Treasure

hydrogen -- Hydrogen is a light gas that makes up most of Jupiter's clouds.

helium -- Helium is the second main gas in Jupiter, the same stuff that makes balloons float.

pressure -- Pressure is the squeezing force that turns Jupiter's gas into liquid the deeper you go.

sulfur -- Sulfur is a yellow mineral that Jupiter's moon Io shoots out in huge volcanic plumes.

radiation -- Radiation is invisible energy that can be very harmful to living things.

Europa -- Europa is one of Jupiter's icy moons that scientists think may hide a giant ocean beneath its surface.

Background you need

Jupiter is the fifth planet from the Sun and has been explored by spacecraft like NASA's Juno mission, which arrived in 2016 to study its clouds and storms.

Europa (one of Jupiter's icy moons) is one of the top places in the solar system where scientists search for signs of alien life.

Jupiter's strong gravity acts like a giant vacuum cleaner, sucking in many asteroids and comets that might otherwise hit Earth.

Break it down

WHO: Jupiter, the largest planet in our solar system

WHAT: A gas giant with wild weather, many moons, and deadly radiation

WHERE: In orbit around the Sun, far beyond Earth

WHY: To show how massive and fascinating Jupiter is, and why it's a place we can only explore with machines

Why it matters

Learning about Jupiter helps us understand how planets form, and one day finding life on its moons could change how you see your place in the universe.

Test what you remember

1. How many Earths could fit inside Jupiter?

- ☐ (A) 13
- ☐ (B) 130
- ☐ (C) 1,300
- ☐ (D) 13,000

2. What are Jupiter's clouds mostly made of?

- ☐ (A) Water droplets
- ☐ (B) Hydrogen and helium
- ☐ (C) Carbon dioxide
- ☐ (D) Oxygen

3. What is the Great Red Spot?

- ☐ (A) A volcano
- ☐ (B) A big lake
- ☐ (C) A giant hurricane
- ☐ (D) A moon

4. How many moons does Jupiter have?

- ☐ (A) 10
- ☐ (B) 35
- ☐ (C) Almost 70
- ☐ (D) Over 100

5. What makes Europa special?

- ☐ (A) It has active volcanoes
- ☐ (B) It has a huge ocean under ice
- ☐ (C) It is the largest moon
- ☐ (D) It has rings

6. Why can't humans land on Jupiter?

- ☐ (A) It is too hot
- ☐ (B) It is too cold
- ☐ (C) It has no solid surface and deadly radiation
- ☐ (D) It is too far away

Answer key (try the quiz first!): 1: C 2: B 3: C 4: C 5: B 6: C

Questions to think about

1. **Curious explorer:** Jupiter's moons might hide alien life, so we want to study them with robotic spacecraft.
2. **Cautious astronaut:** The dangerous radiation around Jupiter would be deadly for people, so humans can't go there--only machines survive.
3. **Weather watcher:** Studying the Great Red Spot teaches us how giant storms can last for centuries, helping us understand weather on other worlds.

MY THOUGHTS (at least 20 words):